

1. Prove that $2^{n+2} + 3^{2n+1}$ is divisible by 7 for every $n \geq 0$.

Solution: First we check the base case: when $n = 0$, we have $2^{0+2} + 3^{2 \cdot 0 + 1} = 2^2 + 3^1 = 4 + 3 = 7$, which indeed is divisible by seven. Now, assume for induction that the statement is true for some n . Let us write

$$2^{(n+1)+2} + 3^{2(n+1)+1} = 2 \cdot 2^{n+2} + 9 \cdot 3^{2n+1} = 2(2^{n+2} + 3^{2n+1}) + 7 \cdot 3^{2n+1}.$$

The first term is divisible by seven by our inductive hypothesis; the second is clearly a multiple of seven. We conclude $2^{(n+1)+2} + 3^{2(n+1)+1}$ is divisible by seven, as desired. This completes the induction.

2. Consider the Fibonacci sequence $(1, 1, 2, 3, 5, 8, 13, \dots)$; that is, let $(f_1, f_2, f_3, \dots)$ be the sequence defined by $f_1 = 1$, $f_2 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 3$.
 1. Prove by induction that every fifth element of the sequence is divisible by five; i.e. prove that $5 \mid f_{5n}$ for each $n \geq 1$.
 2. Prove by induction that $f_1 + f_2 + f_3 + \dots + f_n = f_{n+2} - 1$ for all $n \geq 1$.
 3. Prove by induction that $f_1^2 + f_2^2 + \dots + f_n^2 = f_n f_{n+1}$ for all $n \geq 1$.

Solution:

1. The base case, $n = 1$, is clearly true since $f_5 = 5$. Now assume for induction that f_{5n} is divisible by five. We can write

$$\begin{aligned} f_{5(n+1)} &= f_{5n+5} = f_{5n+4} + f_{5n+3} \\ &= (f_{5n+3} + f_{5n+2}) + (f_{5n+2} + f_{5n+1}) \\ &= ((f_{5n+2} + f_{5n+1}) + (f_{5n+1} + f_{5n})) + ((f_{5n+1} + f_{5n}) + f_{5n+1}) \\ &= (((f_{5n+1} + f_{5n}) + f_{5n+1}) + (f_{5n+1} + f_{5n})) + ((f_{5n+1} + f_{5n}) + f_{5n+1}) \\ &= 5f_{5n+1} + 3f_{5n} \end{aligned}$$

Both of these terms are divisible by 5 (the first is an integer multiple of 5; the second by our inductive hypothesis). So $f_{5(n+1)}$ is divisible by 5, completing our induction.

2. To check the base case, we see $f_1 = 1 = 2 - 1 = f_3 - 1$. Now suppose the statement is true for some n . We write

$$f_1 + f_2 + \dots + f_n + f_{n+1} = (f_{n+2} - 1) + f_{n+1} = f_{n+3} - 1,$$

which is what we wanted to prove. This completes the induction.

3. First we check the base case: $f_1^2 = 1^2 = 1 \cdot 1 = f_1 \cdot f_2$. Now assume the result is true for n . We write

$$f_1^2 + f_2^2 + \dots + f_n^2 + f_{n+1}^2 = f_n f_{n+1} + f_{n+1}^2 = f_{n+1}(f_n + f_{n+1}) = f_{n+1} f_{n+2},$$

which is what we wanted to show. This completes the induction.

3. Prove by induction that, for all $n \in \mathbb{N}$, we have $(1 + 2 + 3 + \cdots + n)^2 = 1^3 + 2^3 + 3^3 + \cdots + n^3$.

Solution: In the base case we have $(1)^2 = 1^3$. Now assume the result is true for some n . We write

$$\begin{aligned}(1 + 2 + 3 + \cdots + n + (n + 1))^2 &= (1 + 2 + \cdots + n)^2 + 2(n + 1)(1 + 2 + 3 + \cdots + n) + (n + 1)^2 \\&= 1^3 + 2^3 + \cdots + n^3 + 2(n + 1)(1 + 2 + 3 + \cdots + n) + (n + 1)^2 \\&= 1^3 + 2^3 + \cdots + n^3 + 2(n + 1)\frac{n(n + 1)}{2} + (n + 1)^2 \\&= 1^3 + 2^3 + \cdots + n^3 + n(n + 1)^2 + (n + 1)^2 \\&= 1^3 + 2^3 + \cdots + n^3 + (n + 1)^3\end{aligned}$$

On the third line we have used the fact that $1 + 2 + \cdots + n = n(n + 1)/2$ (which, if you don't know already, you should prove by induction!). This finishes the inductive argument.